KAIZEN MATHS

Lesson starter

Large Data Set Practice

Context

Name _____

SA1

The data set contains a variable called **hours of sunshine**. State a suitable unit and say whether the variable is categorical, discrete, or continuous.

SA2

Use the row from the Weather & Climate Data. Write one sentence interpreting the rainfall and sunshine values in context.

Town	Month	Year	Mean temp	Rainfall	Sunshine	Wind	Humidity
Asterford	March	2024	9.8 C	52 mm	118 h	21 km/h	78%

Show your working

KAIZEN MATHS

Lesson starter

Large Data Set Practice

Context

Name _____

SB1

A month has no recorded value for hours of sunshine. Explain the problem and suggest a sensible action before analysis.

SB2

The data set contains a variable called **mean temperature**. State a suitable unit and say whether the variable is categorical, discrete, or continuous.

Show your working

KAIZEN MATHS

Exit ticket

Large Data Set Practice

Context

Name _____

EA1

Use the row from the Weather & Climate Data. Write one sentence interpreting the rainfall and sunshine values in context.

Town	Month	Year	Mean temp	Rainfall	Sunshine	Wind	Humidity
Brimley	June	2024	17.9 C	31 mm	196 h	14 km/h	63%

STA-986D-L1 · EA

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KAIZEN MATHS

Exit ticket

Large Data Set Practice

Context

Name _____

EB1

A rainfall value is recorded as -12 mm. Explain the problem and suggest a sensible action before analysis.

STA-986D-L1 · EB

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KAIZEN MATHS

Exit ticket

Large Data Set Practice

Context

Name _____

EC1

The data set contains a variable called **wind speed**. State a suitable unit and say whether the variable is categorical, discrete, or continuous.

STA-986D-L1 · EC

Show your working

KAIZEN MATHS

Exit ticket

Large Data Set Practice

Context

Name _____

EA1

Use the row from the Weather & Climate Data. Write one sentence interpreting the rainfall and sunshine values in context.

Town	Month	Year	Mean temp	Rainfall	Sunshine	Wind	Humidity
Brimley	June	2024	17.9 C	31 mm	196 h	14 km/h	63%

STA-986D-L1 · EA

Show your working

Teacher answers & working

Large Data Set Practice · Context

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SA1

The data set contains a variable called **hours of sunshine**. State a suitable unit and say whether the variable is categorical, discrete, or continuous.

Answer:

hours; continuous

Working:

Identify what is being measured: hours of sunshine.

Use the data dictionary to state the unit: hours.

Classify the variable from how it is measured: continuous.

SA2

Use the row from the Weather & Climate Data. Write one sentence interpreting the rainfall and sunshine values in context.

Town	Month	Year	Mean temp	Rainfall	Sunshine	Wind	Humidity
Asterford	March	2024	9.8 C	52 mm	118 h	21 km/h	78%

Answer:

52 mm rainfall; 118 hours sunshine

Working:

The row is for Asterford in March 2024.

Rainfall is 52 mm, so the town recorded 52 mm of rain that month.

Sunshine is 118 hours, so there were 118 recorded sunshine hours.

Final sentence: In March 2024, Asterford recorded 52 mm of rainfall and 118 hours of sunshine.

SB1

A month has no recorded value for hours of sunshine. Explain the problem and suggest a sensible action before analysis.

Answer:

This is a missing value.

Working:

Identify the type of data issue.

This is a missing value.

Do not treat it as zero; decide whether to omit the row or use a justified imputation method.

A clear cleaning decision should be recorded so later conclusions are transparent.

Teacher answers & working

Large Data Set Practice · Context

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SB2

The data set contains a variable called **mean temperature**. State a suitable unit and say whether the variable is categorical, discrete, or continuous.

Answer:

degrees Celsius; continuous

Working:

Identify what is being measured: mean temperature.

Use the data dictionary to state the unit: degrees Celsius.

Classify the variable from how it is measured: continuous.

EA1

Use the row from the Weather & Climate Data. Write one sentence interpreting the rainfall and sunshine values in context.

Town	Month	Year	Mean temp	Rainfall	Sunshine	Wind	Humidity
Brimley	June	2024	17.9 C	31 mm	196 h	14 km/h	63%

Answer:

31 mm rainfall; 196 hours sunshine

Working:

The row is for Brimley in June 2024.

Rainfall is 31 mm, so the town recorded 31 mm of rain that month.

Sunshine is 196 hours, so there were 196 recorded sunshine hours.

Final sentence: In June 2024, Brimley recorded 31 mm of rainfall and 196 hours of sunshine.

EB1

A rainfall value is recorded as -12 mm. Explain the problem and suggest a sensible action before analysis.

Answer:

This is impossible; rainfall cannot be negative.

Working:

Identify the type of data issue.

This is impossible; rainfall cannot be negative.

Check the original source or mark the value as an error before analysis.

A clear cleaning decision should be recorded so later conclusions are transparent.

Teacher answers & working

Large Data Set Practice · Context

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EC1

The data set contains a variable called **wind speed**. State a suitable unit and say whether the variable is categorical, discrete, or continuous.

Answer:

kilometres per hour; continuous

Working:

Identify what is being measured: wind speed.

Use the data dictionary to state the unit: kilometres per hour.

Classify the variable from how it is measured: continuous.